

Loops (while)

Intro to loop

Loop is one of fundamental concepts of programming.

It is a **control structure** that allows one to repeatedly execute a block of codes.

Components of a loop:

- 1) **Initialization:** one or more variable(s) to keep track of the loop.
- 2) **Condition:** boolean expression based on which the loop runs.
- 3) **Body:** block of codes which is executed each time the loop runs.
- 4) **Steps:** updating the initialized variable(s) to make sure the loop runs certain amount of times only.

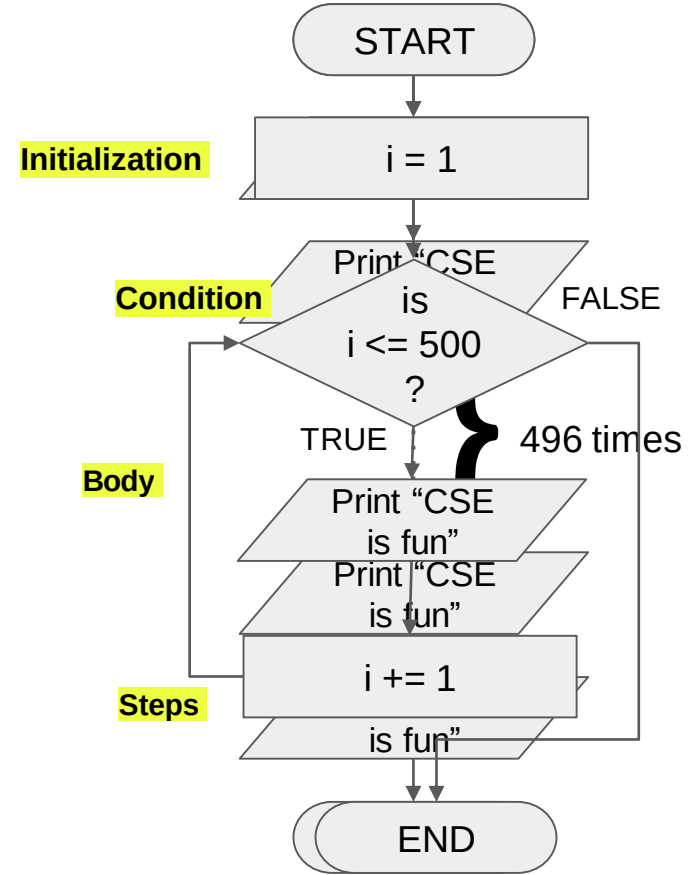
Scenario to Flowchart

Before diving into programming, let's try to visualize how loop actually works using Flowcharts.

Scenario 1: Draw a flowchart that print "CSE is fun" 500 times.

Two Possible solutions:

1. Draw the **Print** block 500 times.
2. Using Loops!!

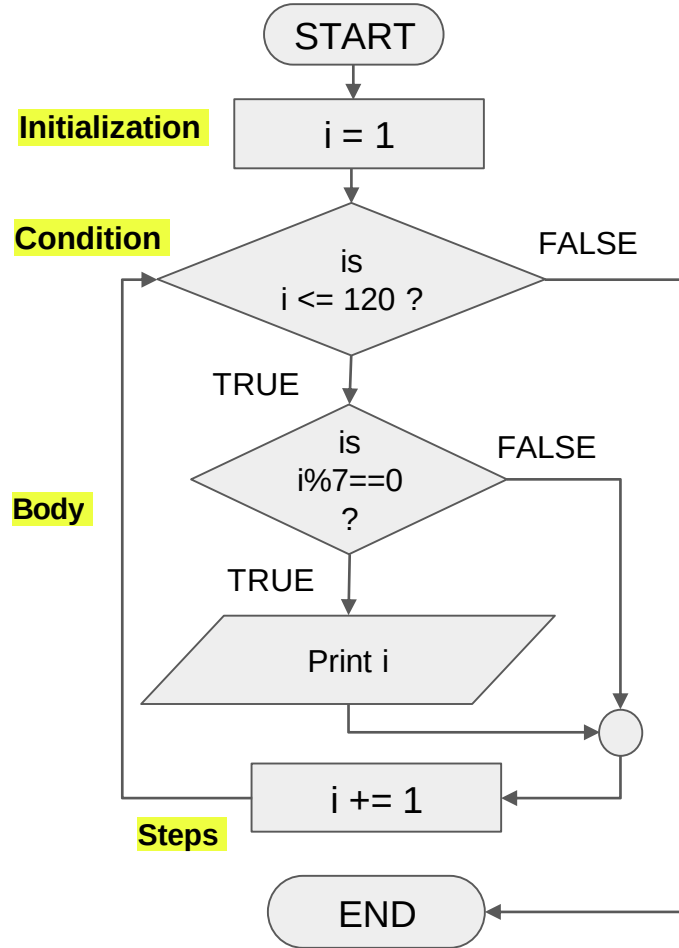


Scenario to Flowchart (Contd)

Let's try to solve another scenario using flowchart

Scenario: Draw a flowchart that prints all numbers between 1 to 120 (including) that are divisible by 7.

As you can see from this example, the looping body can contain branching conditions as well. In complex operations, we can use loops within loops when needed.



Loops in Java

Now that you have a brief visualization of how loops work. Let's head into coding in Java. There are three types of loops in Java.

1. while loop.
2. for loop.
3. do while loop.

- ❖ **for and while** loop only enters the loop's body if the condition is true.
- ❖ On the other hand, **do while** loop executes the body's code first then checks the condition.

while loop

Structure of **while** loop:

1. **INITIALIZATION**
2. **while** (**CONDITION**){
3. **BODY OF THE LOOP**
4. **STEPS**
5. }

Key points of the structure:

- We **initialize** the variable outside the loop's body.
- After the **while** keyword we write the **condition** within parenthesis.
- Then at the end of the loop's body we **increase/decrease** the initialized variable.

while loop (Contd)

Let's try to compare the Scenario 2 flowchart with the Java code of the same problem.

```
1. public class WhileLoopScenario1{
2.     public static void main(String[] args) {
3.         int i = 0;
4.         while ( i < 500 ) {
5.             System.out.println("CSE IS
6.             FUN");
7.             i+=1;
8.         }
9.     }
```

Welcome to DrJava. Working directory is /home/Java110

> run WhileLoopScenario1

CSE IS FUN

CSE IS FUN

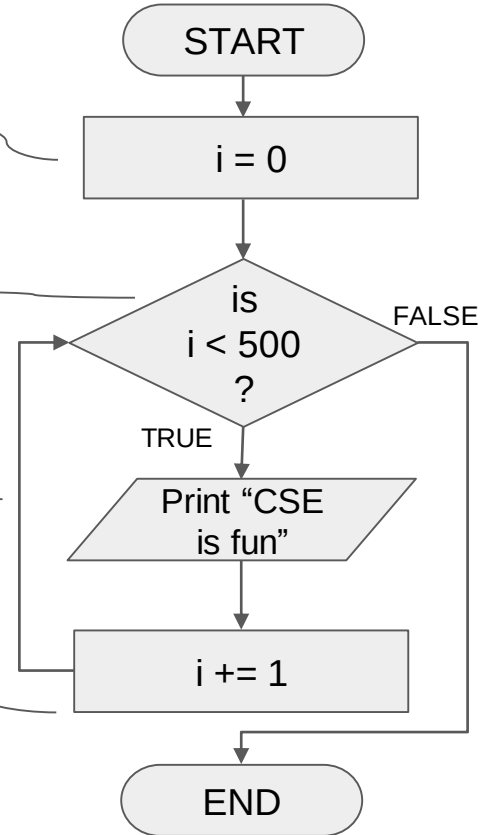
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Initialization

Condition

Body

Steps



while loop (Contd)

How about printing numbers? Let's try to print 1 to 15 using while loop.

```
1. public class WhileLoopNumbers{
2.     public static void main(String[] args) {
3.         int i = 1;
4.         while ( i <= 15 ) {
5.             System.out.print(i + " ");
6.             i += 1;
7.         }
8.     }
9. }
```

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> run WhileLoopNumbers

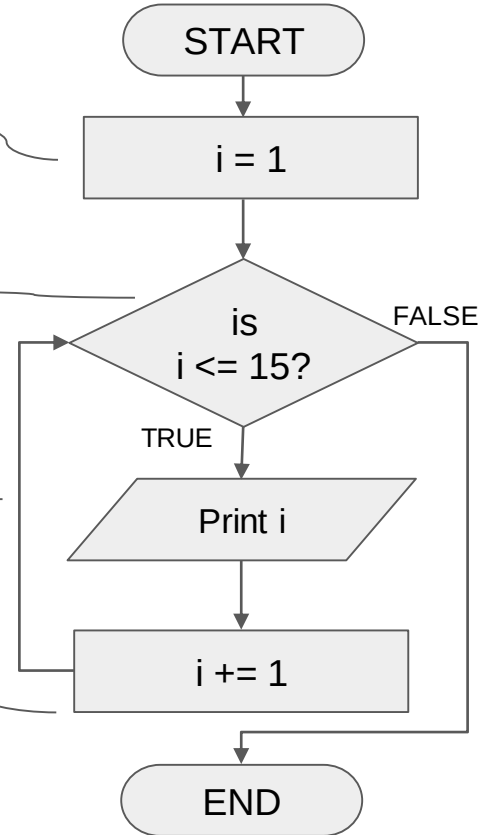
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 >

Initialization

Condition

Body

Steps



while loop (Contd)

Let's try to compare the Scenario 2 flowchart with the Java code of the same problem.

```
1. public class WhileLoopScenario2{
2.     public static void main(String[] args) {
3.         int i = 1;
4.         while ( i <= 120 ) {
5.             if ( i%7==0 ) {
6.                 System.out.print(i+" ");
7.             }
8.             i+=1;
9.         }
10.    }
11. }
```

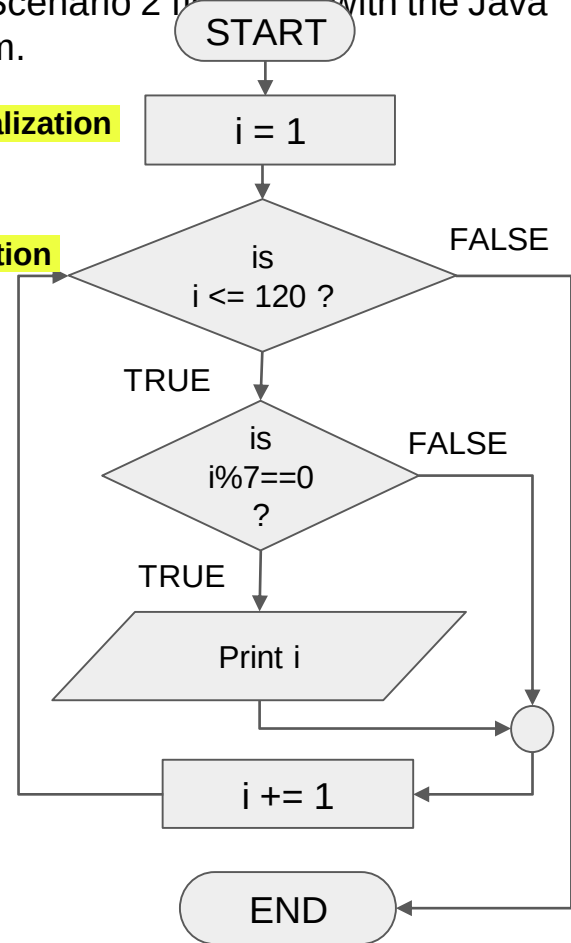
```
Welcome to DrJava. Working directory is /home/Java110
> run WhileLoopScenario2
7 14 21 28 35 42 49 56 63 70 77 84 91 98 105 112 119 >
```

Initialization

Condition

Body

Steps



THANK YOU!